

AquaSonic WP5 State Machine Rocket

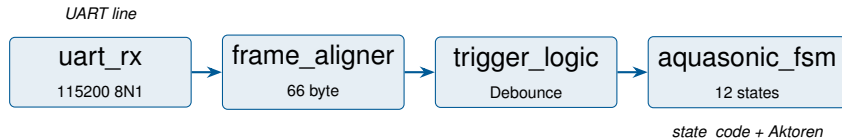
Aktueller Stand

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Was gebaut wurde

Vollständige Sensor-zu-FSM Pipeline in VHDL



Module (6 RTL + 3 TB)

- ▶ aquasonic_pkg – Single Source of Truth
- ▶ uart_rx – 8N1, Mid-Bit-Sampling
- ▶ frame_aligner – Idle-Sync, Field-Extract
- ▶ trigger_logic – 5 Trigger mit Debounce
- ▶ aquasonic_fsm – 3-Prozess, latch-frei
- ▶ rocket_top – Verdrahtung

Highlights

- ▶ Keine Floats / kein sqrt im RTL
- ▶ Alle Schwellwerte in einer Datei
- ▶ Spec-konforme 4-bit State-Codes
- ▶ Self-checking Testbenches

Tests - alle drei Simulationen sauber

Test	Was wird geprüft	Status
sim_fsm.do	12 States, 4 Test-Gruppen	OK
sim_trigger.do	5 Trigger + Debounce	OK
sim_top.do	End-to-End mit echter Flugdaten-CSV	OK

FSM Transcript

```
Transcript:
# Time: 640 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] OVERRIDE -> IDLE
# Time: 670 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] IDLE -> FUELING
# Time: 700 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] FUELING -> OVERRIDE
# Time: 730 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] OVERRIDE -> IDLE
# Time: 760 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] PRE-ARM
# Time: 810 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] PRE-ARM -> OVERRIDE
# Time: 840 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] OVERRIDE -> IDLE
# Time: 870 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: ---- Test 4: critical fault ----
# Time: 870 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] in flight
# Time: 1040 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] fault -> ERROR
# Time: 1070 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: [ OK ] ERROR terminal
# Time: 1100 ns Iteration: 0 Instance: /rocket_fsm_tb
# ** Note: === aquasonic_fsm_tb - DONE ===
# Time: 1100 ns Iteration: 0 Instance: /rocket_fsm_tb
# 0 ps
# 1165500 ps
VSIM 3>
VSIM 3>
```

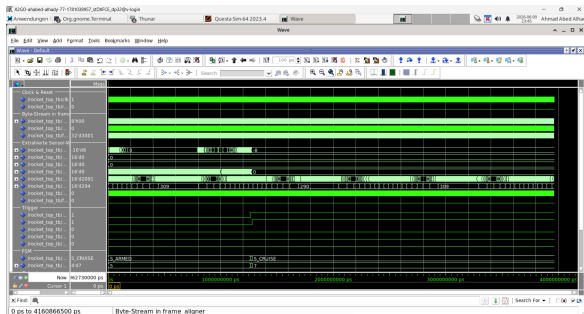
Integration mit CSV

```
Transcript:
# Loading work <sequence>_p1g1000p1
# Loading work.rocket_top_tb(sim)#1
# Loading work.frame_aligner(rtl)#1
# Loading work.trigger_logic(behavioral)#1
# Loading work.aquasonic_fsm(behavioral)#1
# ** Note: === rocket_top_tb - START ===
# Time: 0 ps Iteration: 0 Instance: /rocket_top_tb
# ** Warning: NUMERIC_STD.TO_INTEGER: metavalue detected, returning 0
# Time: 0 ps Iteration: 0 Instance: /rocket_top_tb
# ** Note: STATE -> ERROR
# Time: 0 ps Iteration: 0 Instance: /rocket_top_tb
# ** Note: STATE -> INIT
# Time: 0 ps Iteration: 1 Instance: /rocket_top_tb
# ** Note: STATE -> IDLE
# Time: 95 ns Iteration: 2 Instance: /rocket_top_tb
# ** Note: STATE -> FUELING
# Time: 175 ns Iteration: 2 Instance: /rocket_top_tb
# ** Note: STATE -> PRE_ARM
# Time: 255 ns Iteration: 2 Instance: /rocket_top_tb
# ** Note: STATE -> ARMED
# Time: 335 ns Iteration: 2 Instance: /rocket_top_tb
# ** Note: Pipeline armiert - starte CSV-Stream
# Time: 400 ns Iteration: 0 Instance: /rocket_top_tb
# ** Note: STATE -> LAUNCHED
# Time: 1262335 ns Iteration: 2 Instance: /rocket_top_tb
# ** Note: STATE -> CRUISE
# Time: 1279495 ns Iteration: 2 Instance: /rocket_top_tb
# ** Note: CSV durchgespielt - 3001 Frames gesendet
# Time: 3961720 ns Iteration: 0 Instance: /rocket_top_tb
# ** Note: Letzter State-Code = 7
# Time: 3961720 ns Iteration: 0 Instance: /rocket_top_tb
# ** Note: === rocket_top_tb - DONE ===
# Time: 3962720 ns Iteration: 0 Instance: /rocket_top_tb
# 0 ps

Now: 3,962,730 ns Delta: 1 sim:/rocket_top_tb
```

Detailprotokoll: docs/TEST_LOG.md

Was die echte CSV liefert - und was fehlt



```
sim_top.do Wave-Ansicht
```

Ergebnis aus 3001 Frames

- ▶ Pipeline läuft bis **CRUISE** (Code 7)
- ▶ Liftoff feuert bei Zeile 954
- ▶ Burnout feuert bei Zeile 967
- ▶ Apogee / Main / Landed nicht

Warum?

CSV endet ab Zeile ~964 mit
baro_h = 0 - Sensor-Ausfall im
Originaldatensatz. **Kein RTL-Bug.**

Was wir brauchen

Kompletter Flugdatensatz mit Apogee, Recovery und Touchdown vom Rocket-Team.

Stand & nächste Schritte

Erledigt

- ▶ Komplette Pipeline implementiert
- ▶ Drei Testbenches funktionsfähig
- ▶ Integrationstest mit Echtdaten
- ▶ Konstanten zentralisiert (Package)
- ▶ Doku: README, ARCHITECTURE, TEST_LOG

Offen

- ▶ Kompletter Flugdatensatz für Apogee-Landed Test
- ▶ Schwellwerte final vom Rocket-Team (Tank TBD)
- ▶ Vivado-Synthese auf Trenz Z7020
- ▶ Operator-Command-Decoder mit AquaCom-Team
- ▶ telemetry_tx-Modul (state_code zurück in Paket)

Status: VHDL-Implementierung funktional verifiziert und abgabe-bereit.
Bereit für Synthese sobald Schwellwerte final sind.